Practical No.: 7

Title: Applying Routing in ReactJS

Aim:

To demonstrate **routing in ReactJS** using the **React Router library**, which enables navigation between different pages without reloading the browser.

Theory:

What is React Router?

React Router is a **library for handling navigation** in React applications. It allows the creation of **single-page applications (SPAs)** by enabling navigation between components without full-page reloads.

Concepts Covered:

- **Installing React Router**
- **Creating multiple pages**
- **Using `BrowserRouter`, `Routes`, and `Route`**

Syntax of React Router:

```
import { BrowserRouter, Routes, Route, Link } from "react-router-dom";

<BrowserRouter>
    <nav>
        <Link to="/">Home</Link>
        <Link to="/about">About</Link>
    </nav>
    <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/about" element={<About />} />
    </Routes>
</BrowserRouter>
```

ReactJS Code

Step 1: Create `index.html`

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>React Routing</title>
    <script src="https://unpkg.com/react@18/umd/react.development.js"></script>
    <script src="https://unpkg.com/react-dom@18/umd/react-dom.development.js"></script>
    <script src="https://unpkg.com/babel-standalone@6/babel.min.js"></script>
    <script src="https://unpkg.com/react-router-dom/umd/react-router-dom.min.js"></script>
</head>
<body>
    <h2>Practical 7: Applying Routing</h2>
    <div id="root"></div>
    <script type="text/babel" src="routing.js"></script>
</body>
</html>
```

Step 2: Create index.js

```
const { BrowserRouter, Routes, Route, Link } = ReactRouterDOM;

function Home() {
    return <h3>Welcome to Home Page</h3>;
}

function About() {
    return <h3>This is the About Page</h3>;
}

function App() {
    return (
        <BrowserRouter>
            <nav>
                <Link to="/">Home</Link> |
                <Link to="/about">About</Link>
            </nav>
            <Routes>
                <Route path="/" element={<Home />} />
                <Route path="/about" element={<About />} />
            </Routes>
        </BrowserRouter>
    );
}

const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<App />);
```